

IOT EVENT 2026

SECTION 4

Interactive Knowledge Challenge

Total Marks	20 Marks (24 with Bonus)
Time Allowed	25 Minutes
Question Types	Arrangement Matching Sequencing Mapping
Note	This section is the decider for Round 2 qualification

QUESTION 1: IoT Architecture Layer Arrangement**(4 Marks)**

Arrange the IoT layers in correct order from top (user level) to bottom (physical device level).

Position	Layer Name	Answer
1 (Top)	_____	
2	_____	
3	_____	
4 (Bottom)	_____	

Layers to arrange: Application Layer | Middleware Layer | Network Layer | Perception Layer

ANSWER KEY — Question 1

Position	Layer Name	Scoring
1 (Top)	Application Layer	1 mark
2	Middleware Layer	1 mark
3	Network Layer	1 mark
4 (Bottom)	Perception Layer	1 mark

Scoring: 1 mark for each correctly placed layer = 4 marks total

QUESTION 2: Component Function Matching**(4 Marks)**

Match each component to its correct primary function. Write the letter of the function next to each component.

Component	Function (Write Letter)
1. DHT22	_____
2. MQ-135	_____
3. Relay Module	_____
4. Servo Motor	_____
5. PIR Sensor	_____
6. Ultrasonic Sensor	_____
7. ESP32	_____
8. Breadboard	_____

Functions — choose from below:

Letter	Function
A	Measures temperature and humidity
B	Detects gas, smoke, or air quality
C	Switches high-voltage devices using low-voltage signal
D	Rotates to specific angles for precise movement
E	Detects motion using infrared
F	Measures distance using sound waves
G	Microcontroller with built-in Wi-Fi & Bluetooth
H	Temporary platform for circuit prototyping

ANSWER KEY — Question 2

Component	Answer	Function
1. DHT22	A	Measures temperature and humidity
2. MQ-135	B	Detects gas, smoke, or air quality
3. Relay Module	C	Switches high-voltage devices using low-voltage signal
4. Servo Motor	D	Rotates to specific angles for precise movement
5. PIR Sensor	E	Detects motion using infrared
6. Ultrasonic Sensor	F	Measures distance using sound waves
7. ESP32	G	Microcontroller with built-in Wi-Fi & Bluetooth
8. Breadboard	H	Temporary platform for circuit prototyping

Scoring: 0.5 mark per correct match = 4 marks total

QUESTION 3: Component Interface & Connection Type**(4 Marks)**

Match each component to the correct interface/connection type it uses to communicate with the microcontroller.

Component	Interface Type (Write Letter)
1. LED	_____
2. DHT22	_____
3. Servo Motor	_____
4. MQ-135 Gas Sensor	_____
5. Push Button	_____
6. HC-SR04 Ultrasonic Sensor	_____
7. 16x2 LCD Display (with I2C)	_____
8. PIR Sensor	_____

Interface Types — choose from below:

Letter	Interface Type
A	Digital Output
B	OneWire / Digital Input
C	Analog Output (PWM)
D	Analog Input
E	Digital Input
F	Trigger-Echo (GPIO)
G	I2C
H	Digital Input

ANSWER KEY — Question 3

Component	Ans	Interface Type	Explanation
1. LED	A	Digital Output	Simple ON/OFF control
2. DHT22	B	OneWire / Digital Input	Uses single data wire protocol
3. Servo Motor	C	Analog Output (PWM)	Uses PWM signal for position control
4. MQ-135 Gas Sensor	D	Analog Input	Outputs variable voltage based on gas
5. Push Button	E	Digital Input	Simple HIGH/LOW input
6. HC-SR04	F	Trigger-Echo (GPIO)	Uses TRIG (out) + ECHO (in) pins
7. 16x2 LCD (I2C)	G	I2C	Uses SDA and SCL pins
8. PIR Sensor	H	Digital Input	Outputs HIGH when motion detected

Scoring: 0.5 mark per correct match = 4 marks total

QUESTION 4: Code Logic Sequencing**(4 Marks)**

Arrange the code blocks in the correct order for a complete IoT program that: connects to Wi-Fi, reads temperature from DHT22 sensor, and sends data to cloud via HTTP.

Write the correct sequence (letters A–K) in the table below.

Step	Code Block Letter	Code Block
1	_____	_____
2	_____	_____
3	_____	_____
4	_____	_____
5	_____	_____
6	_____	_____
7	_____	_____
8	_____	_____
9	_____	_____
10	_____	_____
11	_____	_____

Jumbled Code Blocks — choose from below:

Letter	Code Block
A	<code>delay(5000);</code>
B	<code>float temperature = dht.readTemperature();</code>
C	<code>Serial.println(temperature);</code>
D	<code>#include <DHT.h></code>
E	<code>if (isnan(temperature)) { Serial.println("Failed"); return; }</code>
F	<code>dht.begin();</code>
G	<code>http.POST(temperature);</code>

H	<code>#define DHTPIN 4</code>
I	<code>#define DHTTYPE DHT22</code>
J	<code>WiFi.begin(ssid, password);</code>
K	<code>DHT dht(DHTPIN, DHTTYPE);</code>

ANSWER KEY — Question 4

Step	Ans	Code Block	Purpose
1	D	<code>#include <DHT.h></code>	Include DHT library
2	H	<code>#define DHTPIN 4</code>	Define sensor pin
3	I	<code>#define DHTTYPE DHT22</code>	Define sensor type
4	K	<code>DHT dht(DHTPIN, DHTTYPE);</code>	Create DHT object
5	J	<code>WiFi.begin(ssid, password);</code>	Connect to Wi-Fi
6	F	<code>dht.begin();</code>	Initialise DHT sensor
7	B	<code>float temperature = dht.readTemperature();</code>	Read temperature value
8	E	<code>if (isnan(temperature)) { ... }</code>	Check for read errors
9	C	<code>Serial.println(temperature);</code>	Print to Serial monitor
10	G	<code>http.POST(temperature);</code>	Send data to cloud
11	A	<code>delay(5000);</code>	Wait before next reading

Scoring Guide:

Marks	Criteria
4 marks	Completely correct sequence — all 11 steps correct
3 marks	1–2 errors
2 marks	3–4 errors
1 mark	5–6 errors

0 marks	More than 6 errors
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QUESTION 5: Sensor-Actuator Scenario Mapping**(4 Marks)**

For each IoT scenario, write the correct sensor and actuator to complete the system.

Scenario	Sensor	Actuator
1. Smart Irrigation System	_____	_____
2. Automatic Street Light	_____	_____
3. Smoke Detection & Alarm	_____	_____
4. Smart Parking System	_____	_____
5. Temperature-Controlled Fan	_____	_____

Sensor Options	Actuator Options
A Soil Moisture Sensor	P Water Pump / Solenoid Valve
B LDR (Light Dependent Resistor)	Q LED / Street Light
C MQ-135 / Smoke Sensor	R Buzzer
D Ultrasonic Sensor	S Servo Motor / Gate
E DHT22 / Temperature Sensor	T DC Fan / Relay Module

ANSWER KEY — Question 5

Scenario	S.	Sensor	A.	Actuator
1. Smart Irrigation System	A	Soil Moisture Sensor	P	Water Pump / Solenoid Valve
2. Automatic Street Light	B	LDR	Q	LED / Street Light
3. Smoke Detection & Alarm	C	MQ-135 / Smoke Sensor	R	Buzzer
4. Smart Parking System	D	Ultrasonic Sensor	S	Servo Motor / Gate
5. Temperature-Controlled Fan	E	DHT22 / Temperature Sensor	T	DC Fan / Relay Module

Scoring: Sensor correct: 0.4 mark | Actuator correct: 0.4 mark | Per scenario: 0.8 marks | Total: 4 marks

QUESTION 6: Protocol & Communication Match — BONUS QUESTION**(4 Bonus Marks)**

Match each use case to the correct communication protocol. Write the letter of the protocol next to each use case.

Use Case	Protocol (Write Letter)
1. Send sensor data to cloud every 2 seconds (low bandwidth, lightweight)	_____
2. Connect 3 sensors (temperature, humidity, gas) to one microcontroller using 2 wires	_____
3. Control LED from smartphone app (one-time command)	_____
4. Short-range communication between two ESP32 modules	_____
5. Connect OLED display to ESP32	_____
6. Stream video from camera module to web server	_____

Protocols — choose from below:

Letter	Protocol
A	MQTT
B	I2C
C	HTTP / REST API
D	UART / Serial
E	Wi-Fi / TCP/IP
F	SPI

ANSWER KEY — Question 6 (Bonus)

Use Case	Ans	Protocol	Reason
1. Send sensor data to cloud (low bandwidth)	A	MQTT	Lightweight publish-subscribe model

2. Connect 3 sensors using 2 wires	B	I2C	Multiple devices on SDA/SCL lines
3. Control LED from smartphone app	C	HTTP / REST API	Simple request-response model
4. Short-range between two ESP32 modules	D	UART / Serial	Direct device-to-device communication
5. Connect OLED display to ESP32	B	I2C	Standard for displays — uses 2 pins
6. Stream video from camera to web server	E	Wi-Fi / TCP/IP	High bandwidth required

Scoring: 0.67 mark per correct match = 4 bonus marks total

ANSWER KEY SUMMARY

Q	Type	Total Marks	Scoring Method
Q1	Layer Arrangement	4 Marks	1 mark per correct position
Q2	Component Matching	4 Marks	0.5 mark per correct match
Q3	Interface Type Matching	4 Marks	0.5 mark per correct match
Q4	Code Sequencing	4 Marks	Tiered scoring (see Q4 guide)
Q5	Scenario Mapping	4 Marks	0.8 mark per scenario
Q6	Protocol Match (Bonus)	4 Bonus Marks	0.67 mark per correct match
TOTAL		20 Marks (24 with Bonus)	